

SHASHANK KATIYAR

github.com/shashkat [linkedin.com/in/shashkat](https://www.linkedin.com/in/shashkat) shashank_katiyar@dfci.harvard.edu

EDUCATION

Carnegie Mellon University

Master of Science in Computational Biology

GPA: 4.10/4.33

Pittsburgh, PA

December 2024

Indian Institute of Technology Kanpur

Bachelor of Technology in BioSciences and Bioengineering

GPA: 9.1/10 – Department Rank: 1

Kanpur, India

June 2023

SKILLS

Programming: Python, R, C/C++ , Golang, Shell Scripting (UNIX/Linux)

ML Techniques: Random Forest, Gradient Boosting, CNN, Reinforcement Learning, Transformers

Bioinformatics: Single Cell Data Analysis (Scanpy, Seurat), Spatial Data Analysis (spatialdata, squidpy)

Toolkit: Pytorch, Pandas, Numpy, Matplotlib, Seaborn, Git/Github, Docker, AWS, SLURM, HPC

EXPERIENCE

Computational Biologist

Filbin Lab, Dana Farber Cancer Institute

Boston, MA

Jan 2025 – Present

- Performed extensive benchmarking and built a reproducible spatial transcriptomics (10x Xenium) analysis pipeline to support multiple projects, improving analysis efficiency and scalability across projects
- Led analysis and optimization of spatial dataset for a Nature-published study, implementing a novel "coherence" metric to quantify tumor organization and reveal recurrence-associated spatial patterns
- Developed custom computational workflows integrating spatial transcriptomics with immunofluorescence (image) and calcium imaging data (video) to uncover celltype-specific molecular and functional patterns
- Developed a pipeline for segmentation, quantification and subsequent visualization of multichannel IF (immunofluorescence) images. Packaged it using hatch project manager for efficient collaboration
- Currently working on integrating a spatial transcriptomics dataset composed of different subtypes of pediatric high grade gliomas to discover spatial patterns in the biology of these tumors

Computational Biology Intern

Novasenta, Inc.

Pittsburgh, PA

May 2024 – August 2024

- Identified celltype abundances in Lung Cancer Visium samples using cell2location spot deconvolution, improving company's understanding of localization of target ligand-receptor (L-R) pairs in tumor tissue
- Inferred celltype-celltype interactions using Linear NCEM (Node Centric Expression Models) in spot deconvoluted samples gaining further evidence of the L-R interaction between expected celltypes
- Quantified L-R colocalization in VisiumHD data with binomial test; got significant colocalization, $p < 0.01$
- Implemented Nuclei Segmentation on VisiumHD data to get spatial-single cell data for L-R analysis and found multiple celltype-pair interactions with pvalue < 0.05 according to custom-made permutation test

Machine Learning Intern

TenSixty BioSciences, Inc.

Cambridge, MA

Sep 2022 – July 2023

- Developed a scalable pipeline for rare cell marker identification in single cell data, including containerized implementation on AWS-parallel for high-throughput isoform-level alignment
- Identified a differentially expressed isoform in small cell lung cancer for targeted drug development and designed a high-accuracy protein binder using Rfdiffusion
- Developed 3D visualization Python script, for identification of extracellular binding sites of Protein

OPEN SOURCE CONTRIBUTION

Squidpy - Spatial Single Cell Analysis in Python

Code reviewer and contributor

March 2026 – Present

- Contributed to the open-source Python package Squidpy by identifying and resolving bugs, improving code stability, and enhancing overall functionality through targeted pull requests
- Regularly collaborate with maintainers and contributors to review code, shape feature roadmaps, and implement new analytical capabilities for spatial transcriptomics workflows

VOLUNTEERING

Turn the Bus - online education for underprivileged students

Volunteer for biology tutoring

October 2025 – Present

- Mentor tutors to develop educational content for eleventh standard biology in India for underprivileged students. Attend weekly progress meetings with fellow volunteers

PROJECTS

Single Cell Analysis of Engineered AAVs for Gene Therapy

Pittsburgh, PA

Mentor: Prof. Leah Byrne, University of Pittsburgh

Jan 2024– May 2024

- Implemented Slurm scripts to run Cellranger on HPC cluster on scRNAseq FASTQs from primate retina
- Performed quality control followed by celltype annotation on processed scRNAseq files using Scanpy
- Identified presence of different adeno-associated virus (AAV) barcodes in cells using Salmon tool
- Mapped AAV infection patterns to celltypes, revealing preferences of AAV variants for cell populations, helping identify potential candidates for targeted gene delivery strategies

PUBLICATIONS

- Jeong, D., Danielli, S.G., Maaß, K.K., David R. Ghasemi, Svenja K. Tetzlaff, Ekin Reyhan, Li Jiang, **Shashank Katiyar**, ... Kristian W. Pajtler, Varun Venkataramani, Mariella G. Filbin. Multidimensional profiling of heterogeneity in supratentorial ependymomas. **Nature** 652, 1016–1026 (2026).
- Jędrzej Kubica, Rachit Kumar, Glenda Tan, Van Q. Truong, David Enoma, Nicholas Cooley, Minhyek Jeon, Chiao-Feng Lin, Minh Tran, Amrita Roy Choudhury, Xinrong Du, **Shashank Katiyar**, et al. The fourth annual Carnegie Mellon Libraries hackathon for biomedical data management, knowledge graphs, and deep learning. **BioRxiv** (2023).